

Detailed Answer Key

1 FILL-IN CONCEPTUAL CHECKS

Concepts needed: field definition; conductor equilibrium; Gauss law; traveling waves; SHM; refraction; polarization; sound; potential slope; ions

These blanks test vocabulary and the direction of the physical relationships. The important habit is to name the object being described before writing an equation: E belongs to the source arrangement, flux counts net field through a closed boundary, and a wave function describes a moving pattern. Sign and direction are part of the concept, not decorations. For example, the minus sign in SHM says the acceleration restores the equilibrium position, while the minus sign in $F = -dU/dr$ says force points toward decreasing potential energy. Similarly, chlorine becomes negative because it gains an electron, whereas ionization of an atom removes an electron and leaves a positive ion.

- 1: force; charge. 2: zero. 3: enclosed. 4: $x-vt$. 5: opposite. 6: speed. 7: volume. 8: longitudinal. 9: negative first. 10: Cl^- ; gain.
- For the wave blank, $x-vt$ gives a right-moving pattern: keeping $f(x-vt)$ fixed requires x to increase as t increases.
- For the potential blank, differentiate with respect to separation and reverse the sign; the slope, not the height alone, determines force.

FULL WORKED RELATIONS / CHECKS

$$E = F/q$$

$$\oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}}/\epsilon_0$$

$$y = f(x-vt)$$

$$F = -dU/dr$$

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2 MATCHING VISUALS TO EQUATIONS

Concepts needed: Gauss symmetry; translation form of a wave; potential-energy slope; dielectric constitutive definition

Matching is a representation problem: identify what a picture is emphasizing before scanning the equation list. A closed Gaussian sphere and a point charge highlight enclosed charge and symmetry, so Gauss's law is the match. A translated sinusoid is described by a function of $x-vt$. A potential well is read through its derivative because force is the negative slope. Aligned microscopic dipoles are material polarization, so the displacement definition combines the physical field with P . The same strategy works on unfamiliar diagrams: ask what is constant, what changes, and whether the diagram is geometric, kinematic, energetic, or material-response based.

1. A-1: the spherical surface makes E constant and radial, reducing flux to EA .
2. B-2: $x-vt$ keeps the shape fixed while translating it to the right.
3. C-3: the minimum has zero slope, so force is zero there; the curve's slope changes sign around it.
4. D-4: P summarizes dipoles per volume and $D=\epsilon_0 E+P$ accounts for the dielectric response.

FULL WORKED RELATIONS / CHECKS

$$\mathbf{A} \rightarrow 1; \mathbf{B} \rightarrow 2; \mathbf{C} \rightarrow 3; \mathbf{D} \rightarrow 4$$

$$\Phi = EA \text{ when } E \text{ is uniform and normal}$$

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$$

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3 SHM GRAPH AND QUANTITIES

Concepts needed: Hooke law; angular frequency; period; phase; energy conservation; derivative phase shifts

The cosine form starts at maximum positive displacement, so $x(0)=A$ and $v(0)=0$. Differentiation shifts the phase by a quarter cycle: $v(t)=-A\omega \sin(\omega t)$, meaning velocity initially becomes negative as the mass returns toward equilibrium. At $x=+A$ the acceleration is most negative because the restoring force is largest, while speed is zero. The period is the time for the phase to change by 2π . Energy gives speed at any position without needing to solve for time, because spring and kinetic energy exchange while their sum stays constant in ideal SHM.

1. With $A=0.20$ m and $\omega=4.00$ rad/s: $T=2\pi/\omega=1.57$ s.
2. $v_{\max}=A\omega=(0.20)(4.00)=0.800$ m/s.
3. At $x=+A$: $a=-\omega^2 A=-(4.00)^2(0.20)=-3.20$ m/s² and $v=0$.
4. Graph A is $x(t)$; Graph B can be $v(t)$ because $v(0)=0$ and v initially decreases for $\phi=0$.
5. $\frac{1}{2}mv^2+\frac{1}{2}kx^2=\frac{1}{2}kA^2$, so $v(x)=\pm\sqrt{[(k/m)(A^2-x^2)]}$.

FULL WORKED RELATIONS / CHECKS

$$\begin{aligned}x &= A \cos(\omega t) \\ v &= dx/dt = -A\omega \sin(\omega t) \\ a &= dv/dt = -\omega^2 x \\ T &= 2\pi/\omega\end{aligned}$$

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4 TRAVELING-WAVE CALCULATION

Concepts needed: amplitude; wave number; angular frequency; wavelength; frequency; period; phase; wave speed; partial derivative

For a sinusoidal wave, compare the given expression term-by-term with $y=A \cos(kx-\omega t)$. The coefficient outside the cosine is amplitude, the coefficient of x is wave number, and the coefficient of t is angular frequency. The negative sign before ωt means rightward translation. Convert angular quantities using 2π , then use $v=\omega/k$ or $v=f\lambda$. The requested transverse velocity is a partial derivative: x is held fixed while differentiating with respect to time. This distinguishes particle motion from the propagation speed of the pattern.

1. $A=0.040$ m; $k=6.00$ rad/m; $\omega=120$ rad/s.
2. $\lambda=2\pi/k=1.047$ m; $f=\omega/(2\pi)=19.10$ Hz; $T=1/f=0.0524$ s.
3. $v=\omega/k=20.0$ m/s. Since the phase is $kx-\omega t$, the wave travels in $+x$.
4. $v_y=dy/dt=0.040(120)\sin(6x-120t)$. At $x=t=0$, $v_y=0$.
5. Units check: k is rad/m, ω is rad/s, and their ratio is m/s.

FULL WORKED RELATIONS / CHECKS

$$y=0.040 \cos(6.00x-120t) \text{ m}$$

$$\lambda=2\pi/6.00=1.047 \text{ m}$$

$$f=120/(2\pi)=19.10 \text{ Hz}$$

$$v=120/6.00=20.0 \text{ m/s}$$

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5 CONSTANT-ACCELERATION DERIVATIONS

Concepts needed: derivatives; definite integrals; initial conditions; integration constants; kinematics

These derivations show why the familiar kinematics formulas are consequences of the differential definitions, not independent facts. The first derivation starts with acceleration and uses limits to encode the initial velocity. The second integrates the resulting velocity and uses the initial position. Differentiating the final expressions is the strongest check: it returns the original velocity and acceleration exactly.

1. Since $dv/dt=a$, multiply by dt : $dv=a dt$. Integrate from v_i at $t=0$ to v at t : $v-v_i=\int_0^t a dt'=at$.
2. Therefore $v(t)=v_i+at$ and at $t=t_f$, $v_f=v_i+at_f$. The constant v_i is required by $v(0)=v_i$.
3. Next use $dx/dt=v_i+at$. Integrate from x_i at $t=0$: $x-x_i=\int_0^t (v_i+at')dt'=v_i t+1/2at^2$.
4. Hence $x(t)=x_i+v_i t+1/2at^2$. Differentiating once returns v_i+at ; differentiating twice returns a .

FULL WORKED RELATIONS / CHECKS

$$\begin{aligned}
 dv/dt &= a \\
 v_f - v_i &= \int_0^t a dt' = at \\
 v_f &= v_i + at \\
 x_f - x_i &= \int_0^t (v_i + at') dt' = v_i t + 1/2 at^2 \\
 x_f &= x_i + v_i t + 1/2 at^2
 \end{aligned}$$

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6 GAUSS LAW PROOF AND SPHERE

Concepts needed: inverse-square field; flux; spherical symmetry; enclosed charge; volume integration

Gauss's law is especially powerful when symmetry makes the field magnitude constant on a chosen closed surface. The proof begins with the point-charge field and uses the area of a sphere. The application then changes only the enclosed charge according to the region. Outside a uniform sphere, the full charge is enclosed and the result is point-charge-like. Inside, the enclosed charge scales with r^3 while the Gaussian area scales with r^2 , producing a field proportional to r . An arbitrary shape lacks a surface on which E is both constant and normal, so Gauss's law remains true but is no longer a one-line solver.

1. For a sphere: $\Phi = \int \mathbf{E} \cdot d\mathbf{A} = E(4\pi r^2) = (kQ/r^2)(4\pi r^2) = 4\pi kQ = Q/\epsilon_0$.
2. Outside ($r \geq R$): $Q_{\text{enc}} = \rho(4\pi R^3/3)$, so $E = Q_{\text{enc}}/(4\pi\epsilon_0 r^2) = \rho R^3/(3\epsilon_0 r^2)$.
3. Inside ($r < R$): $Q_{\text{enc}} = \rho(4\pi r^3/3)$, so $E(4\pi r^2) = Q_{\text{enc}}/\epsilon_0$ and $E = \rho r/(3\epsilon_0)$.
4. At $r=R$ both expressions agree: $E = \rho R/(3\epsilon_0)$, providing a boundary check.

FULL WORKED RELATIONS / CHECKS

$$\begin{aligned}
 k &= 1/(4\pi\epsilon_0) \\
 \int \mathbf{E} \cdot d\mathbf{A} &= Q_{\text{enc}}/\epsilon_0 \\
 E_{\text{inside}} &= \rho r/(3\epsilon_0) \\
 E_{\text{outside}} &= \rho R^3/(3\epsilon_0 r^2)
 \end{aligned}$$

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7 FINITE ROD: TRIG SUBSTITUTION

Concepts needed: continuous charge; superposition; symmetry; $dq = \lambda dx$; component resolution; trigonometric substitution

The field from a continuous rod is built from small charge elements. Symmetry eliminates the horizontal components pair-by-pair, leaving only the vertical component. The trigonometric substitution is useful because $r = \sqrt{a^2 + x^2}$ appears in both the inverse-square field and the component factor. Let $x = a \tan \theta$: then $r = a \sec \theta$, $dx = a \sec^2 \theta d\theta$, and $\cos \theta = a/r = \cos \theta$. The integral becomes simple and the limits encode the rod endpoints. This is the general pattern to reuse: write dq , write the distance, resolve the component, substitute, then use symmetry.

1. $dE = k dq/r^2 = k\lambda dx/(a^2 + x^2)$. The vertical component is $dE_y = dE \cos \theta$, with $\cos \theta = a/r$.
2. Thus $dE_y = k\lambda a dx/(a^2 + x^2)^{3/2}$. Integrate x from $-L$ to L .
3. Using $x = a \tan \theta$: $dx = a \sec^2 \theta d\theta$ and $(a^2 + x^2)^{3/2} = a^3 \sec^3 \theta$, so $dE_y = (k\lambda/a) \cos \theta d\theta$.
4. The limits are $\theta = \pm \tan^{-1}(L/a)$. Therefore $E_y = (k\lambda/a) [\sin \theta]_{-\theta_0}^{\theta_0} = (2k\lambda/a) \sin \theta_0$.
5. Since $\sin \theta_0 = L/\sqrt{a^2 + L^2}$, $E = 2k\lambda L/[a\sqrt{a^2 + L^2}]$. As $L \rightarrow \infty$, $E \rightarrow 2k\lambda/a = \lambda/(2\pi\epsilon_0 a)$.

FULL WORKED RELATIONS / CHECKS

$$E = 2k\lambda L / (a\sqrt{a^2 + L^2})$$

$$L \rightarrow \infty: E = 2k\lambda/a = \lambda/(2\pi\epsilon_0 a)$$

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8 POTENTIAL-ENERGY GRAPH

Concepts needed: Lennard-Jones model; power-law terms; derivative; stability; point-charge comparison

The graph is determined by competing powers of separation. At very small r , the positive r^{-12} term dominates and produces a steep repulsive wall. At intermediate r , the negative r^{-6} term can dominate, making U negative and creating an attractive well. At large r both terms vanish, so U approaches the chosen zero. Force depends on slope: the minimum is not special because U is negative; it is special because the slope is zero and the curvature is positive, making the equilibrium stable to small radial displacements.

1. Differentiate term by term: $dU/dr = -12C_{12}/r^{13} + 6C/r^7$.
2. Therefore $F = -dU/dr = 12C_{12}/r^{13} - 6C/r^7$.
3. Equilibrium requires $F=0$, so $12C_{12}/r^{13} = 6C/r^7$ and $r_6 = 2C_{12}/C$.
4. The minimum is stable because the curve bends upward there: a small displacement creates a restoring force.
5. The point-charge potential kq_1q_2/r has no short-range repulsive wall by itself; realistic ions and molecules require extra structure.

FULL WORKED RELATIONS / CHECKS

$$U = C_{12}/r^{12} - C/r^6$$

$$F = 12C_{12}/r^{13} - 6C/r^7$$

$$r_{eq} = (2C_{12}/C)^{1/6}$$

Detailed Answer Key

9 ATOMIC MODELS AND DIELECTRIC DERIVATION

Concepts needed: Bohr model; electron cloud; ionization; valence electrons; dipole moment; polarization; susceptibility; permittivity

The atomic portion tests model selection. Bohr orbits are a useful quantized picture for simple atoms, but the electron-cloud model is more accurate because it describes probability rather than a definite track. Ion formation follows electron counting: losing negative charge makes an ion positive, while gaining it makes an ion negative. The dielectric derivation then moves from microscopic to macroscopic description. A dipole moment describes one separated pair; polarization averages many dipoles per volume; D packages the free-charge field and material response. For a linear isotropic medium, P is parallel and proportional to E , which permits the definition of ϵ .

1. Bohr force balance: $ke^2/r^2 = m_e v^2/r$. Quantum restrictions determine allowed states; the force balance alone is classical.
2. Na loses one valence electron $\rightarrow \text{Na}^+$. Cl gains one electron $\rightarrow \text{Cl}^-$. Ionization of a neutral atom removes an electron and leaves a positive ion.
3. $p = qd$ and $P = (1/V)\sum p_i$. Then $D = \epsilon_0 E + P$.
4. Insert $P = \epsilon_0 \chi_e E$: $D = \epsilon_0 E + \epsilon_0 \chi_e E = \epsilon_0 (1 + \chi_e) E$.
5. Define $\epsilon = \epsilon_0 (1 + \chi_e)$, giving $D = \epsilon E$. In the isotropic sketch E , P , and D are collinear.

FULL WORKED RELATIONS / CHECKS

$$\begin{aligned}
 p &= qd \\
 P &= (1/V)\sum p_i \\
 D &= \epsilon_0 E + P \\
 P &= \epsilon_0 \chi_e E \\
 D &= \epsilon_0 (1 + \chi_e) E = \epsilon E
 \end{aligned}$$

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10 SOUND PRESSURE AND SNELL LAW

Concepts needed: rms pressure; intensity; logarithms; decibels; refractive index; phase continuity

The sound calculation uses two linked ideas. Sound-pressure level is defined from pressure relative to a reference, while intensity is proportional to the square of rms pressure. Taking the difference between two levels cancels the reference and leaves a pressure ratio. Refraction is separate but uses the same disciplined ratio reasoning: frequency is continuous across the boundary, so speed and wavelength change. Snell's law expresses tangential phase matching. Since n_2 is larger, the second medium is slower and the ray bends toward the normal.

1. Pressure ratio = $60/20 = 3$, so $dL_p = 20 \log_{10} 3 = 9.54$ dB.
2. Because $I \propto p^2$, $I_2/I_1 = (3)^2 = 9$; equivalently $10 \log_{10} 9 = 19.08$ dB for intensity level, consistent with the pressure-level factor 20.
3. Snell: $1.00 \sin 40^\circ = 1.50 \sin \theta_2$, so $\sin \theta_2 = 0.4285$ and $\theta_2 = 25.4^\circ$.
4. The ray bends toward the normal because $\theta_2 < \theta_1$. The frequency stays fixed; v and λ decrease in the larger-index medium.
5. A correct drawing includes the interface, normal, incident angle from the normal, and the smaller refracted angle.

FULL WORKED RELATIONS / CHECKS

$$\begin{aligned}
 dL_p &= 20 \log_{10} (p_2/p_1) \\
 I_2/I_1 &= (p_2/p_1)^2 \\
 n_1 \sin \theta_1 &= n_2 \sin \theta_2 \\
 \theta_2 &= \sin^{-1} [(1.00/1.50) \sin 40^\circ] = 25.4^\circ
 \end{aligned}$$

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11 DERIVATION SYNTHESIS

Concepts needed: Hooke law; Newton law; partial derivatives; wave translation; phase continuity; refractive index; linear assumptions

This final item checks whether the equations can be rebuilt from physical laws. For SHM, the crucial assumption is a linear restoring force: Hooke law. For a traveling wave, the crucial assumption is a sinusoidal shape with constant propagation speed, and differentiation reveals that spatial curvature and temporal acceleration have the same shape with a scale factor. For Snell law, the interface must preserve phase along its tangent: the tangential component of the wave vector is continuous. These are not memorization tricks. Each derivation connects a local physical rule to a global equation and exposes the assumptions that would fail for nonlinear springs, dispersive media, or an interface that does not maintain a common frequency.

1. SHM: $F = -kx$ and $F = mx''$ imply $mx'' = -kx$. Divide by m and define $\omega^2 = k/m$: $x'' + \omega^2 x = 0$. The assumption is linear restoring behavior.
2. Wave: $y = A \cos(kx - \omega t)$. Two x derivatives give $y_{xx} = -k^2 y$; two t derivatives give $y_{tt} = -\omega^2 y$. Since $v = \omega/k$, $y_{xx} = (1/v^2)y_{tt}$.
3. The wave equation assumes a constant v and small enough behavior for linear superposition; it is a verification of the proposed solution.
4. Snell: in equal time dt , phase fronts advance $v_1 dt$ and $v_2 dt$. Tangential phase matching gives $v_1 dt \sin\theta_1 = v_2 dt \sin\theta_2$.
5. Cancel dt and substitute $v = c/n$: $\sin\theta_1 / \sin\theta_2 = v_2 / v_1 = n_1 / n_2$, hence $n_1 \sin\theta_1 = n_2 \sin\theta_2$.

FULL WORKED RELATIONS / CHECKS

$$\begin{aligned}
 mx'' &= -kx \rightarrow x'' + \omega^2 x = 0 \\
 y_{xx} &= -k^2 y; y_{tt} = -\omega^2 y \rightarrow y_{xx} = (1/v^2)y_{tt} \\
 v_1 \sin\theta_1 &= v_2 \sin\theta_2 \rightarrow n_1 \sin\theta_1 = n_2 \sin\theta_2
 \end{aligned}$$